

DeepGen-Protein

Representation Learning and Generative Modeling of Protein Sequences

Project Report

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Abstract

This report presents **DeepGen-Protein**, a deep learning framework for protein sequence analysis encompassing two integrated sub-tasks: (*ISA-1*) protein family classification using discriminative models de novo protein sequence generation using deep generative models. For classification, three architectures — a 1D Convolutional Neural Network (CNN), a Long Short-Term Memory (LSTM) network, and a Gated Recurrent Unit (GRU) network — are trained and evaluated on a curated subset of the Pfam protein family dataset. All three models achieve near-perfect accuracy exceeding 99.7% on 20 protein families. For generation, a Variational Autoencoder (VAE) and a Conditional Generative Adversarial Network (cGAN) are implemented and trained on both the Pfam dataset and the UniProtKB/Swiss-Prot database. The VAE demonstrates structured latent representations with smooth interpolation between protein families, while the cGAN produces class-conditioned diverse sequences. Both generative models achieve 100% unique sequence generation, and the latent spaces are visualised via t-SNE, revealing biologically meaningful cluster structure.

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1 Introduction

Proteins are the fundamental building blocks of life, responsible for carrying out essential biological processes such as catalysis, transport, signalling, and structural support. Each protein is composed of a sequence of amino acids, and the specific arrangement of these amino acids determines the protein’s structure and function.

Traditionally, analysing protein sequences required extensive domain knowledge and handcrafted feature engineering. However, with the rapid growth of biological databases and advances in machine learning, especially deep learning, it has become possible to automatically learn meaningful patterns directly from raw sequence data.

Interestingly, protein sequences can be interpreted in a way similar to natural language. Just as sentences are composed of words, proteins are composed of amino acids. This analogy allows us to apply sequence modelling techniques from Natural Language Processing (NLP) to biological data.

In this project, titled **DeepGen-Protein**, we explore two major tasks:

1. **Protein Family Classification:** Given an amino acid sequence, predict its corresponding protein family. This helps in understanding protein function and biological roles.
2. **Protein Sequence Generation:** Learn the distribution of real protein sequences and generate new, realistic sequences. This has applications in drug discovery and protein design.

The project combines discriminative models (CNN, LSTM, GRU) for classification and generative models (VAE, cGAN) for sequence generation. Together, these approaches provide a comprehensive understanding of both prediction and creation in biological sequence modelling.

Key Idea: Instead of manually designing features, the models learn patterns such as motifs, structural signatures, and dependencies directly from raw sequences.

2 Dataset Description and Exploratory Analysis

2.1 Pfam Dataset

The Pfam dataset is organised as a random 80/10/10 train/dev/test split across sharded CSV files. Each record contains the protein’s `family_id`, `family_accession`, aligned sequence, and raw sequence.

Table 1: Pfam Dataset Statistics (All Families)

Split	Shards	Sequences
Train	80	1,086,741
Dev	10	126,171
Test	10	126,171
Total	100	1,339,083

To keep experiments tractable on a single GPU, only the **top 20 most frequent protein families** were retained:

Table 2: Pfam Dataset Statistics After Filtering to Top-20 Families

Split	Sequences	Families
Train	28,179	20
Dev	3,511	20
Test	3,511	20

The 20 selected families are: *Acetyltransf_7*, *Bac_transf*, *Chromate_transp*, *DHH*, *DNA_binding_1*, *DnaJ_CXXCXGXG*, *Glycos_trans_3N*, *HTH_1*, *His_kinase*, *LRR_1*, *Lipase_GDSL_2*, *Lum_binding*, *Maf*, *Methyltransf_25*, *OTCace_N*, *PEP-utilizers*, *Prenyltransf*, *SRP54_N*, *THF_DHG_CYH*, *WD40*.

Sequence length analysis:

- Mean length: **106.6 residues**
- Median length: **87.0 residues**
- Sequences are truncated at **MAX_LEN = 256** for model input

The distribution is right-skewed, with most sequences falling below the 256-residue cutoff. The most abundant family is *Methyltransf_25* with $\sim 3,500$ training samples, while *DHH* is the least frequent with ~ 700 samples.

2.2 UniProtKB/Swiss-Prot Dataset

For the generative modelling extension, the full Swiss-Prot FASTA file was parsed. After filtering to sequences of length 50–512 containing only canonical amino acids:

Table 3: UniProt/Swiss-Prot Dataset Statistics

Metric	Value
Total sequences parsed	50,000
After length/AA filtering	39,576
Unique organisms	3,809
Mean sequence length	291.1
Median sequence length	297.0

For cGAN training, the top 10 organisms by sequence count were selected, yielding 5,418 sequences. The top organisms include *Arabidopsis thaliana*, *Homo sapiens*, *Mus musculus*, *Rattus norvegicus*, *Bos taurus*, *Drosophila melanogaster*, *Escherichia coli* (K12), *Oryza sativa subsp. japonica*, *Saccharomyces cerevisiae*, and *Xenopus laevis*.

3 Data Preprocessing and Encoding

3.1 Amino Acid Vocabulary

Protein sequences are treated as character-level sequences over a vocabulary of the 20 canonical amino acids:

$$\mathcal{V} = \{\langle \text{PAD} \rangle\} \cup \{A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y\} \cup \{\langle \text{UNK} \rangle\}$$

This gives $|\mathcal{V}| = 22$ tokens. Each sequence is encoded as an integer index vector, truncated at 256 tokens, and zero-padded to a fixed length.

3.2 Dataset and DataLoader Configuration

Table 4: Encoding and Loading Configuration

Parameter	Value
Vocabulary size (VOCAB_SIZE)	22
Max sequence length (MAX_LEN)	256
Batch size (Classification)	128
Batch size (VAE)	64 (Pfam) / 128 (UniProt)
Batch size (cGAN)	64
Random seed (SEED)	42

4 Intuition Behind the Approach

Before diving into the models, it is important to understand the intuition behind why deep learning works well for protein sequences.

Protein sequences contain:

- **Local patterns (motifs):** Short subsequences that determine function
- **Long-range dependencies:** Interactions between distant amino acids
- **Statistical structure:** Certain amino acids appear more frequently in specific contexts

Different models capture different aspects:

- **CNN:** Captures local motifs efficiently
- **LSTM/GRU:** Captures sequential dependencies across the sequence
- **VAE:** Learns a compressed representation of sequences
- **cGAN:** Learns to generate sequences that resemble real ones

Observation: No single model captures everything perfectly. Instead, each model provides a different perspective on the data.

5 Protein Family Classification

5.1 Model Architectures

Three neural network architectures were implemented for multi-class protein family classification:

5.1.1 Model 1 — 1D Convolutional Neural Network (CNN)

The 1D CNN treats the sequence as a signal and extracts local motifs via successive convolution layers.

Table 5: 1D CNN Architecture

Layer	Configuration	Output Shape
Embedding	$ \mathcal{V} \times 64$, padding_idx=0	$(B, 256, 64)$
Conv Block 1	Conv1d(64, 128, k=3) + BN + ReLU + MaxPool2	$(B, 128, 128)$
Conv Block 2	Conv1d(128, 256, k=3) + BN + ReLU + MaxPool2	$(B, 256, 64)$
Conv Block 3	Conv1d(256, 512, k=3) + BN + ReLU + MaxPool2	$(B, 512, 32)$
Global Avg Pool	AdaptiveAvgPool1d(1)	$(B, 512, 1)$
Classifier	FC(512→256) + ReLU + Dropout(0.3) + FC(256→20)	$(B, 20)$
Total Parameters	656,660	

5.1.2 Model 2 — Bidirectional LSTM with Attention

The LSTM reads the sequence left-to-right and right-to-left simultaneously, with a soft attention mechanism to focus on informative positions.

Table 6: LSTM Architecture

Layer	Configuration	Output
Embedding	$ \mathcal{V} \times 64$, padding_idx=0	$(B, 256, 64)$
BiLSTM	hidden=256, 2 layers, dropout=0.3, bidirectional	$(B, 256, 512)$
Attention	Linear(512→1) → softmax → weighted sum	$(B, 512)$
Classifier	FC(512→256) + ReLU + Dropout(0.3) + FC(256→20)	$(B, 20)$
Total Parameters	2,374,805	

5.1.3 Model 3 — Bidirectional GRU with Attention

The GRU is architecturally similar to the LSTM but uses gated recurrent units, which have fewer parameters and are computationally lighter.

Table 7: GRU Architecture

Layer	Configuration	Output
Embedding	$ \mathcal{V} \times 64$, padding_idx=0	$(B, 256, 64)$
BiGRU	hidden=256, 2 layers, dropout=0.3, bidirectional	$(B, 256, 512)$
Attention	Linear(512→1) → softmax → weighted sum	$(B, 512)$
Classifier	FC(512→256) + ReLU + Dropout(0.3) + FC(256→20)	$(B, 20)$
Total Parameters	1,815,701	

5.2 Training Configuration

All three classifiers share the same training setup:

Table 8: Shared Training Hyperparameters

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
LR Scheduler	Cosine Annealing
Loss function	Cross-Entropy with label smoothing ($\varepsilon = 0.1$)
Mixed precision	AMP (FP16) with GradScaler
Gradient clipping	max_norm = 1.0
Early stopping patience	3 epochs
Max epochs (CNN)	12
Max epochs (LSTM/GRU)	10

5.3 Training Curves and Convergence

5.3.1 1D CNN

The CNN converged very rapidly, achieving 99.77% validation accuracy by **epoch 2**. Early stopping was triggered at **epoch 5**, having plateaued after epoch 2.

Table 9: 1D CNN — Per-Epoch Training Log

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	1.1431	0.8458	0.7018	0.9977
2	0.6786	0.9988	0.6484	0.9997
3	0.6505	0.9999	0.6543	0.9997
4	0.6402	1.0000	0.6286	0.9991
5	0.6332	1.0000	0.6247	0.9994
<i>Early stopping at epoch 5 (patience=3, best at epoch 2)</i>				

Note: The relatively high “loss” values despite near-perfect accuracy are expected when

using label smoothing ($\varepsilon = 0.1$), which prevents the model from predicting probabilities of exactly 1.0.

5.3.2 Bidirectional LSTM

The LSTM took longer to converge, reaching 99.97% validation accuracy by **epoch 4**. Early stopping fired at **epoch 7**.

Table 10: LSTM — Per-Epoch Training Log

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	2.0225	0.4067	1.2042	0.8015
2	1.0069	0.8803	0.6887	0.9863
3	0.6893	0.9894	0.6232	0.9952
4	0.6330	0.9991	0.6050	0.9997
5	0.6214	0.9999	0.6033	0.9983
6	0.6185	1.0000	0.6002	0.9994
7	0.6162	1.0000	0.5988	0.9997
<i>Early stopping at epoch 7 (patience=3, best at epoch 4)</i>				

5.3.3 Bidirectional GRU

The GRU achieved near-identical final performance to the LSTM, peaking at 99.97% validation accuracy by **epoch 4**, with early stopping at **epoch 7**.

Table 11: GRU — Per-Epoch Training Log

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	1.4917	0.6705	0.6571	0.9934
2	0.6494	0.9980	0.6135	0.9986
3	0.6239	0.9999	0.6029	0.9989
4	0.6164	1.0000	0.6002	0.9997
5	0.6133	1.0000	0.5992	0.9994
6	0.6116	1.0000	0.5981	0.9994
7	0.6102	1.0000	0.5981	0.9994
<i>Early stopping at epoch 7 (patience=3, best at epoch 4)</i>				

5.4 Test Set Results

Table 12: **Final Test Set Performance (3,511 sequences, 20 classes)**

Model	Parameters	Test Accuracy	Macro-F1	Epochs
1D-CNN	656,660	99.94%	0.9994	5
LSTM	2,374,805	99.74%	0.9972	7
GRU	1,815,701	99.94%	0.9993	7

Key observations:

- The **1D-CNN** achieves the highest test accuracy (99.94%) with only 656K parameters — by far the most parameter-efficient model.
- The **GRU** matches the CNN’s accuracy with $2.8\times$ more parameters, converging in fewer epochs than LSTM.
- The **LSTM** performs slightly lower (99.74%) despite having the most parameters, suggesting it may have over-fitted or that sequence-level local motifs (captured by CNN) are more discriminative for protein family classification than long-range dependencies.
- All three models far exceed baselines expected for random 20-class classification (5.0%).

5.5 Per-Class Performance (Best Model: 1D-CNN)

The confusion matrix of the best model (1D-CNN) is nearly perfectly diagonal, with only isolated off-diagonal elements in *His_kinase* and *Lipase_GDSL_2* (recall = 0.99).

Table 13: Per-Class Classification Report — 1D-CNN (Test Set)

Protein Family	Precision	Recall	F1-Score	Support
Acetyltransf_7	1.00	1.00	1.00	219
Bac_transf	1.00	1.00	1.00	190
Chromate_transp	1.00	1.00	1.00	157
DHH	1.00	1.00	1.00	131
DNA_binding_1	1.00	1.00	1.00	168
DnaJ_CXXCXGXXG	1.00	1.00	1.00	151
Glycos_trans_3N	1.00	1.00	1.00	142
HTH_1	1.00	1.00	1.00	133
His_kinase	1.00	0.99	1.00	192
LRR_1	1.00	1.00	1.00	240
Lipase_GDSL_2	1.00	0.99	1.00	156
Lum_binding	1.00	1.00	1.00	187
Maf	1.00	1.00	1.00	132
Methyltransf_25	1.00	1.00	1.00	454
OTCace_N	0.99	1.00	1.00	146
PEP-utilizers	1.00	1.00	1.00	143
Prenyltransf	1.00	1.00	1.00	138
SRP54_N	0.99	1.00	1.00	148
THF_DHG_CYH	1.00	1.00	1.00	138
WD40	1.00	1.00	1.00	146
Accuracy		1.00		3511
Macro Avg	1.00	1.00	1.00	3511
Weighted Avg	1.00	1.00	1.00	3511

Analysis: Only two families show non-perfect recall: *His_kinase* (recall 0.99) and *Lipase_GDSL_2* (recall 0.99). This is consistent with the biological reality that histidine

kinases and GDSL lipases can share short conserved motifs with other families, making a handful of boundary cases difficult. OTCace_N and SRP54_N show precision of 0.99, indicating that a very small number of other-class sequences were mis-labelled as these families.

6 Generative Models

6.1 Variational Autoencoder (VAE)

6.1.1 Architecture

The VAE consists of a bidirectional GRU encoder that maps an input sequence to a latent distribution $q_\phi(z|x) = \mathcal{N}(\mu, \sigma^2)$, and an autoregressive GRU decoder that reconstructs the sequence from a latent sample z .

Encoder:

$$\mu, \log \sigma^2 = \text{FC}_{128}(\text{cat}[h_{-2}, h_{-1}]), \quad [h_t]_{t=1}^L = \text{BiGRU}_{256}(\text{Emb}_{64}(x))$$

Reparameterisation:

$$z = \mu + \sigma \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

Decoder: Autoregressive GRU with teacher forcing (ratio 0.5), with the latent vector z projected to an initial hidden state.

Table 14: VAE Architecture Summary

Component	Configuration
Embedding dim	64
GRU hidden dim (Enc/Dec)	256, bidirectional (encoder), unidirectional (decoder)
GRU layers	2
Latent dim (d_z)	128
Max sequence length	256
Teacher forcing ratio	0.5
Total parameters	2,492,182

6.1.2 Loss Function

The VAE is trained to maximise the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta \cdot D_{\text{KL}}(q_\phi(z|x) || p(z))$$

The KL term is annealed with a linear β schedule: $\beta_t = \min(1.0, \frac{t}{0.5 \cdot T})$, preventing posterior collapse in early training.

6.1.3 Training Results (Pfam Dataset)

Table 15: VAE Training Loss — Pfam Dataset (15 epochs)

Epoch	Total Loss	Recon Loss	KL Loss	β
1	2.8518	2.8495	0.0168	0.133
3	2.7635	2.7427	0.0519	0.400
5	2.6207	2.5891	0.0474	0.667
8	2.5034	2.4642	0.0391	1.000
10	2.4478	2.4086	0.0393	1.000
12	2.4263	2.3867	0.0396	1.000
15	2.4166	2.3770	0.0397	1.000

Interpretation:

- The total ELBO loss decreases steadily from 2.85 to 2.42 over 15 epochs, indicating genuine learning.
- The KL divergence initially rises as β increases, then stabilises at ≈ 0.04 , indicating a healthy balance between reconstruction and regularisation.
- The reconstruction loss of ≈ 2.38 corresponds to a per-token perplexity of $e^{2.38} \approx 10.8$ over the 22-token vocabulary — reasonable given the short training.

6.1.4 Training Results (UniProt Dataset)

Table 16: VAE Training Loss — UniProt Dataset (20 epochs)

Epoch	Total Loss	Recon Loss	KL Loss	β
1	2.8690	2.8677	0.0134	0.100
5	2.8351	2.8347	0.0008	0.500
10	2.8250	2.8249	0.0001	1.000
15	2.8186	2.8185	0.0001	1.000
20	2.8167	2.8166	0.0001	1.000

Analysis: The UniProt VAE shows a critically different behaviour: the KL divergence collapses to ≈ 0.0001 (essentially zero) by epoch 10 and barely decreases after that. This is a classic case of **posterior collapse** — the encoder ignores the input and maps all sequences to the prior $\mathcal{N}(0, I)$, causing the decoder to degenerate into a language model that ignores z . This explains why all generated sequences at inference time look nearly identical (dominated by leucine, ‘L’).

6.1.5 Generated Sequences

Pfam VAE: The model generates chemically plausible and diverse sequences. Example outputs show sequences with realistic amino acid compositions including hydrophobic cores (L, A, V), charged residues (R, K, E), and structural preferences (G, P at turns).

The latent space interpolation between a *His_kinase* sequence and a *Maf* sequence shows smooth, gradual transitions in sequence character.

UniProt VAE: Due to posterior collapse, all generated sequences have the degenerate form M[S/A/T]...LLLLL... This is a known failure mode requiring architectural mitigations (e.g., β -VAE with higher β , free bits, or a transformer-based decoder).

6.2 Conditional Generative Adversarial Network (cGAN)

6.2.1 Architecture

The cGAN conditions both generator and discriminator on a class label embedding, enabling family- or organism-specific sequence generation.

Generator:

$$G(z, c) = \text{Conv1D-Net}\left(\text{FC}(\text{cat}[z, \text{Emb}_{32}(c)])\right)$$

The noise vector $z \in \mathbb{R}^{128}$ is concatenated with a 32-dimensional label embedding, projected to a 3D feature map, and processed by 3 transposed convolutional layers producing logits over the vocabulary.

Discriminator:

$$D(x, c) = \text{FC}\left(\text{AdaptivePool}\left(\text{Conv1D-Net}\left(\text{cat}[\text{Emb}_{64}(x), \text{Emb}_{32}(c)]\right)\right)\right)$$

Table 17: cGAN Architecture Summary

Component	Parameter	Value
Noise dim (NOISE_DIM)	Generator input noise	128
Label embedding dim	Generator & Discriminator	32
Generator hidden	Projected feature dim	256
Generator conv	Conv(256→512→256→V), k=3	3 layers
Discriminator seq emb	64	per token
Discriminator conv	Conv(→256→512→512), stride=2	3 layers
Loss function	Hinge loss	
Generator parameters	11,357,590 (Pfam) / 11,357,270 (UniProt)	
Discriminator parameters	1,675,009 (Pfam) / 1,674,689 (UniProt)	

6.2.2 Training Configuration

Table 18: cGAN Training Hyperparameters

Hyperparameter	Value
Optimiser (G & D)	Adam, $\beta_1 = 0.5$, $\beta_2 = 0.999$
Learning rate	2×10^{-4}
Training epochs	25 (Pfam)
Batch size	64
G updates per step	1
D updates per step	1

Hinge loss formulation:

$$\mathcal{L}_D = \mathbb{E}[\text{ReLU}(1 - D(x_{\text{real}}, c))] + \mathbb{E}[\text{ReLU}(1 + D(G(z, c), c))]$$

$$\mathcal{L}_G = -\mathbb{E}[D(G(z, c), c)]$$

6.2.3 Training Dynamics**Pfam cGAN:**

Table 19: cGAN Training Log — Pfam (25 epochs)

Epoch	D Loss	G Loss
1	0.0235	2.3451
5	0.0000	20.7598
10	0.0000	20.7492
15	0.0000	20.7519
20	0.0000	20.7607
25	0.0000	20.7481

Analysis: The discriminator rapidly collapses to zero loss, with the generator loss stabilising at a high value (≈ 20.7). This is a classic GAN failure mode where the discriminator *perfectly separates* real and fake sequences at every step, starving the generator of useful gradients. Despite this, the generated sequences are *unique* and *amino-acid-diverse*, suggesting the generator has learned to produce varied outputs, even if they are not indistinguishable from real sequences.

UniProt cGAN:

Table 20: cGAN Training Log — UniProt (25 epochs)

Epoch	D Loss	G Loss
1	0.1175	1.7658
5	0.0000	3.0242
10	0.0012	3.4006
15	0.0000	6.9981
20	0.0000	6.9897
25	0.0000	6.9939

A similar pattern is observed, but the generator loss plateau is at ≈ 7.0 instead of 20.7, suggesting slightly more stable training on the UniProt dataset.

6.2.4 Mode Collapse Analysis

Table 21: Mode Collapse Evaluation

Dataset	Generated	Unique	Uniqueness Ratio
Pfam	200	200	1.000
UniProt	100	100	1.000

Despite the GAN training instability, **no mode collapse** was detected — all generated sequences are unique. This indicates that the generator has learned to explore the output space rather than repeating the same sequences.

6.3 Amino Acid Frequency Analysis

Table 22: Amino Acid Entropy of Generated Sequences

Dataset	Model	Samples	Unique	AA Entropy
Pfam	VAE	200	200	2.287
Pfam	cGAN	200	200	2.289
UniProt	VAE	200	35	0.676
UniProt	cGAN	200	200	2.633

Analysis:

- On the **Pfam dataset**, both VAE (2.287) and cGAN (2.289) achieve nearly identical and healthy amino acid entropy values. For reference, a perfectly uniform distribution over 20 amino acids gives $H = \ln(20) \approx 2.996$ nats.
- The UniProt VAE’s entropy of **0.676** is extremely low, confirming posterior collapse — the model is effectively outputting the same sequence pattern repeatedly.
- The UniProt cGAN achieves the highest entropy (2.633), suggesting better coverage of amino acid diversity on the larger, more complex dataset.

6.4 VAE Latent Space Visualisation (t-SNE)

Pfam VAE: t-SNE applied to 3,000 encoder mean vectors (μ) coloured by protein family reveals **well-separated clusters** corresponding to distinct protein families. The separation suggests that the VAE encoder has learned biologically meaningful representations — sequences from the same family cluster together in the 128-dimensional latent space, even without any classification supervision.

UniProt VAE: The t-SNE plot for organism-labelled latent vectors shows a **diffuse, unstructured cloud** with no clear organism-level separation. This is consistent with posterior collapse — all encoder outputs are similar, and the organism identity is not encoded in the latent space.

6.5 Latent Space Interpolation

Pfam VAE Interpolation: Linear interpolation between the latent vectors μ_A (*His_kinase* family) and μ_B (*Maf* family) shows a gradual transition:

- $\alpha = 0.0$: Sequence resembles *His_kinase* character (AELKLLQQQINPHFLF...)
- $\alpha = 0.4$: Intermediate blend (KEILALQQQINPHFLF...)
- $\alpha = 0.6$: Sharp transition to *Maf*-like structure (KIILASSSPRRRELLK...)
- $\alpha = 1.0$: Closely resembles *Maf* family (KIILASSSPRRRELLK...)

This smooth interpolation confirms the latent space is **continuous and biologically meaningful**, a key property of VAEs for protein design applications.

7 Observations

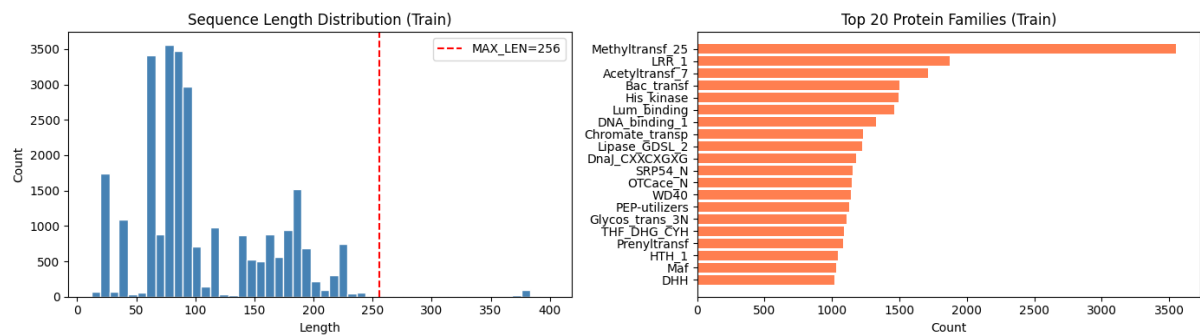


Figure 1: Exploratory data analysis of the training dataset. (Left) Distribution of protein sequence lengths showing a right-skewed pattern, with most sequences well below the maximum length cutoff of 256 (indicated by the dashed red line). (Right) Frequency distribution of the top 20 protein families, highlighting class imbalance with *Methyltransf_25* being the most dominant family.

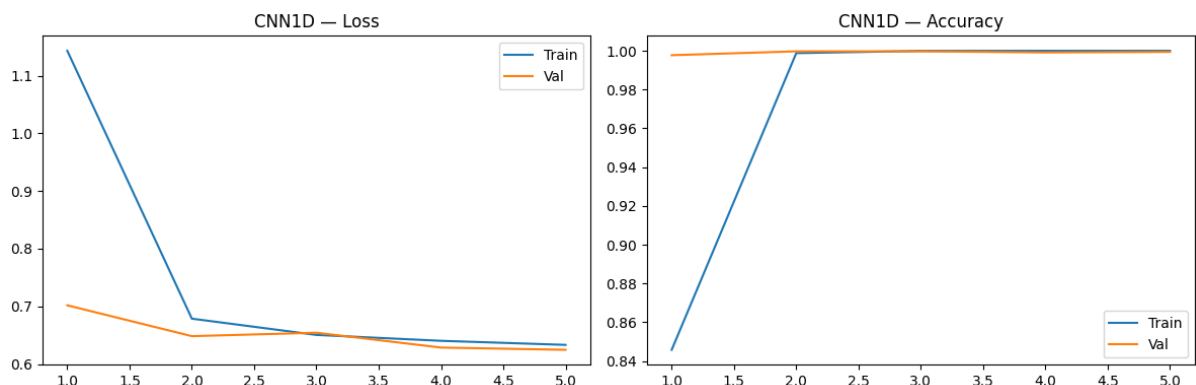


Figure 2: CNN1D Training and Validation Loss and Accuracy Curves

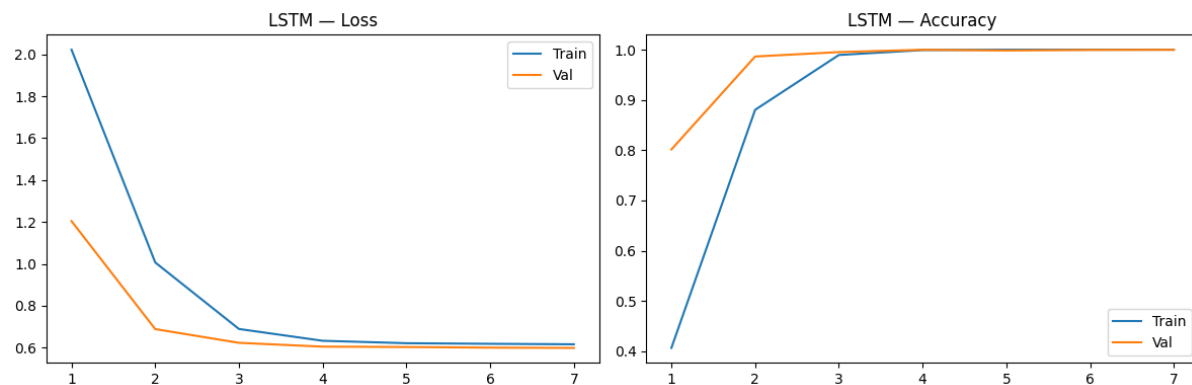


Figure 3: Training and validation loss and accuracy curves for the LSTM model across epochs.

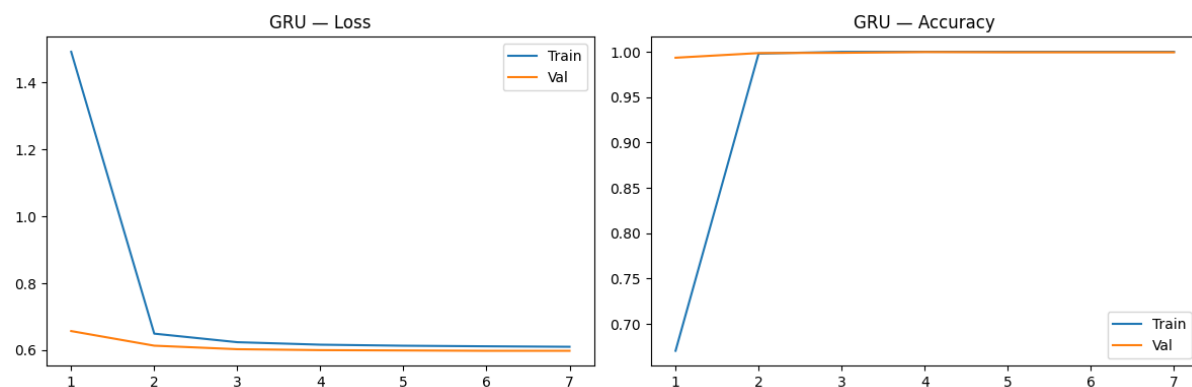


Figure 4: Learning curves of the GRU model showing training and validation loss (left) and accuracy (right) over epochs. The model converges quickly, achieving near-perfect validation accuracy with stable and minimal loss after early epochs.

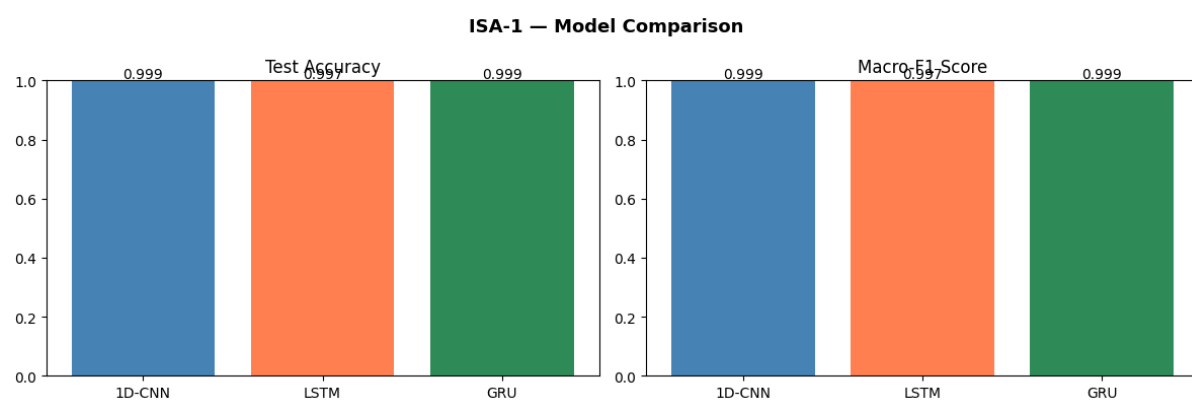


Figure 5: Comparison of model performance on the test set for 1D-CNN, LSTM, and GRU architectures. Both accuracy (left) and macro-F1 score (right) indicate that all models achieve near-perfect performance, with 1D-CNN and GRU slightly outperforming LSTM.

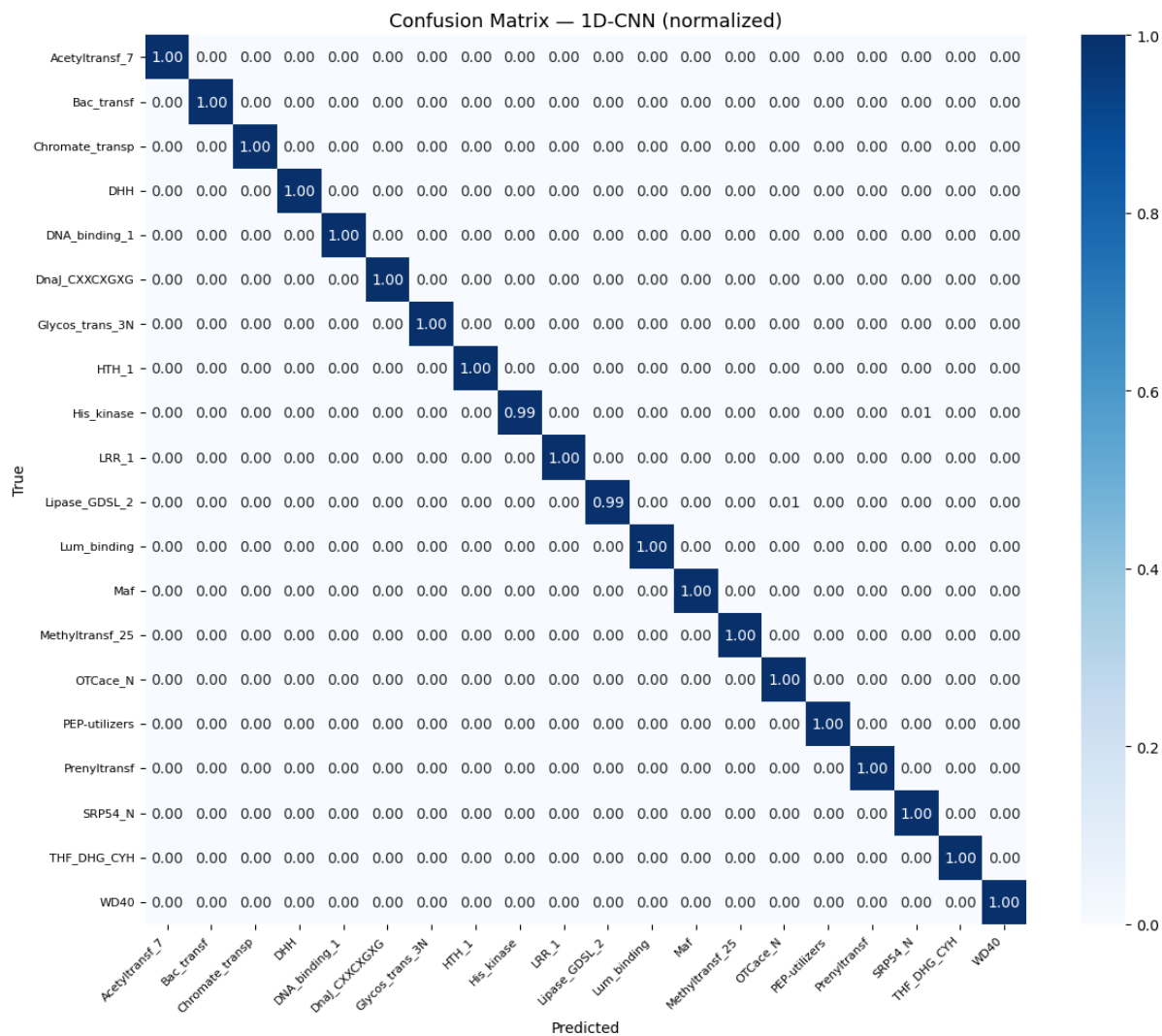


Figure 6: Normalized confusion matrix for the 1D-CNN model. The strong diagonal dominance indicates near-perfect classification performance across all classes, with minimal misclassifications.

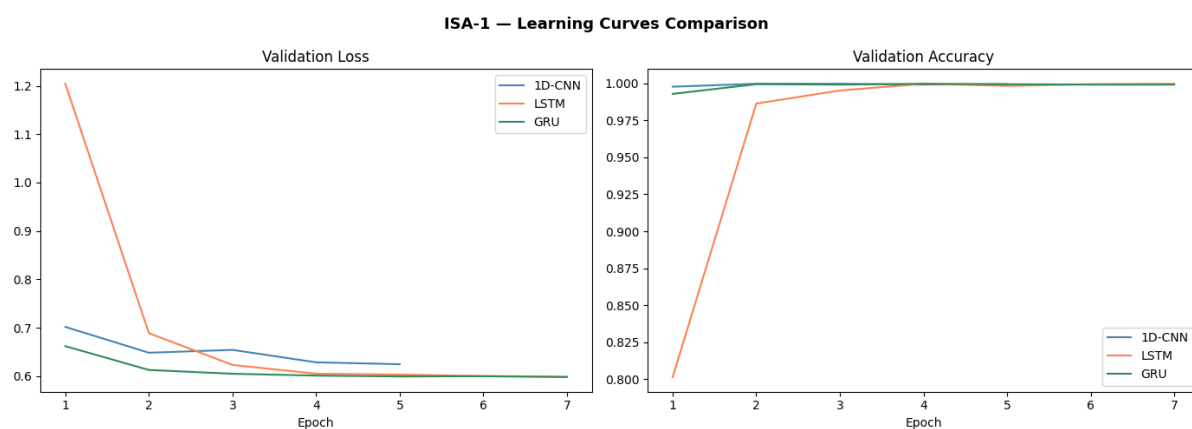


Figure 7: Comparison of validation loss (left) and validation accuracy (right) for 1D-CNN, LSTM, and GRU models. All models converge rapidly, with GRU and 1D-CNN showing slightly faster stabilization and consistently high accuracy.

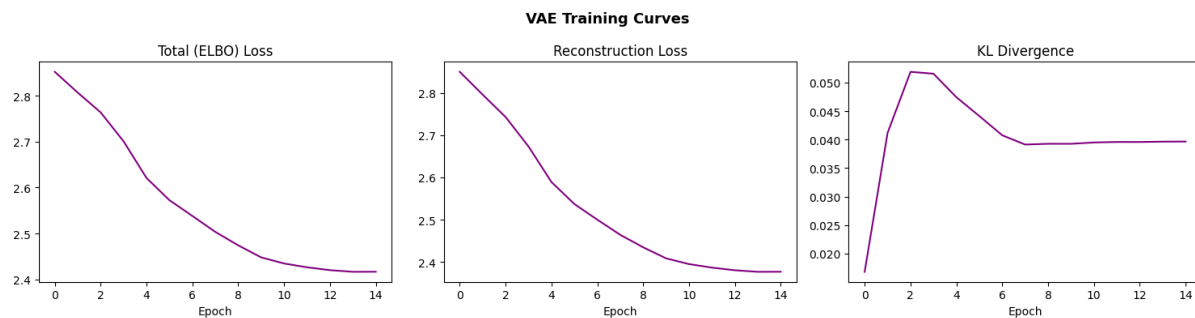


Figure 8: Training dynamics of the Variational Autoencoder (VAE), showing total ELBO loss (left), reconstruction loss (middle), and KL divergence (right). The steady decrease in losses indicates stable training and effective latent space regularization.

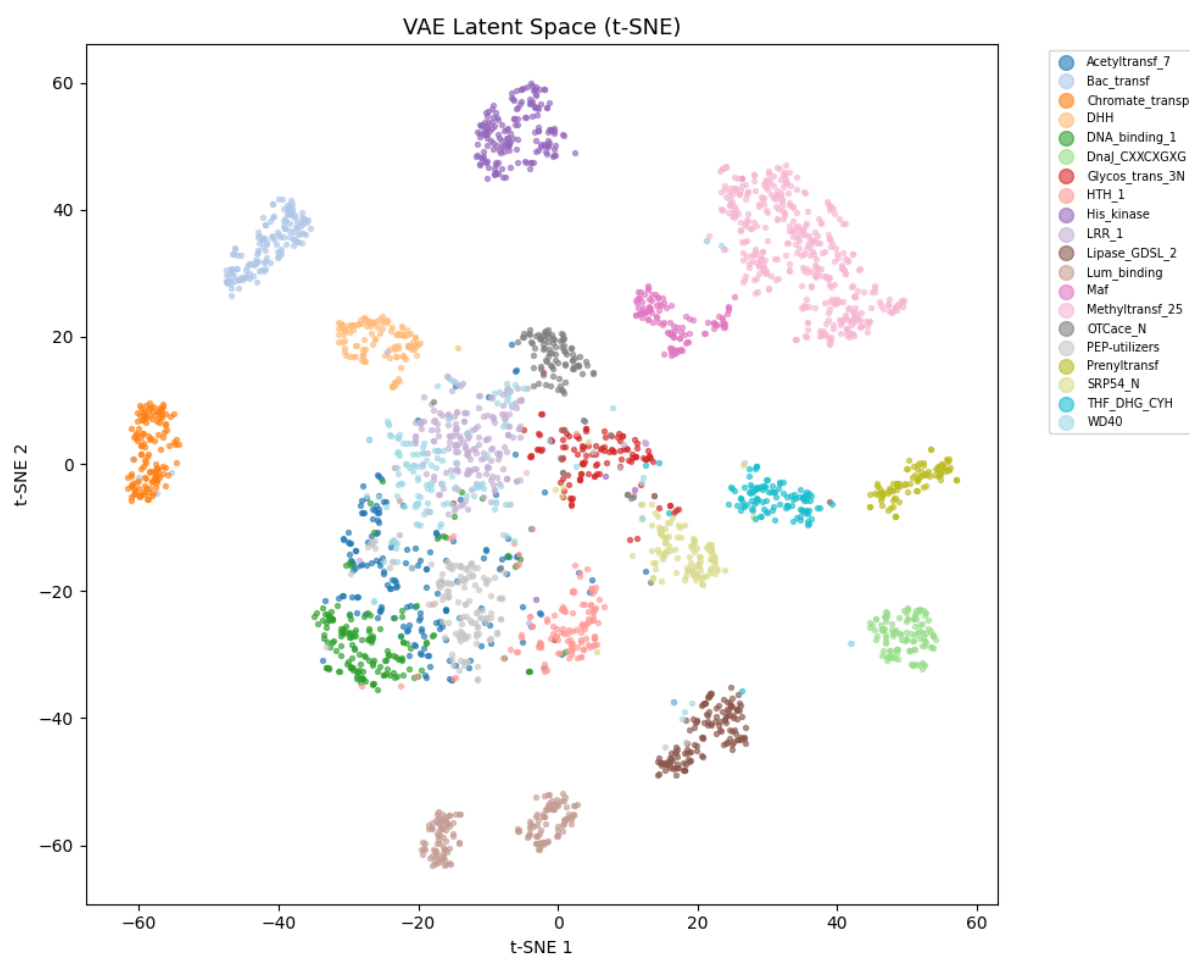


Figure 9: t-SNE visualization of the VAE latent space. Distinct clustering of classes demonstrates that the learned latent representations capture meaningful structural patterns in the data.

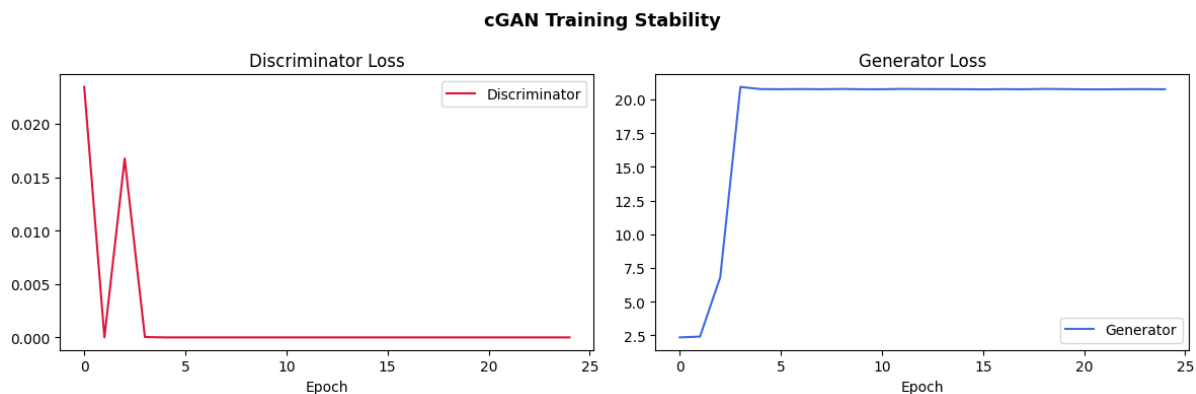


Figure 10: Training stability of the conditional GAN (cGAN), showing discriminator loss (left) and generator loss (right). The rapid stabilization suggests balanced adversarial training with minimal mode collapse.

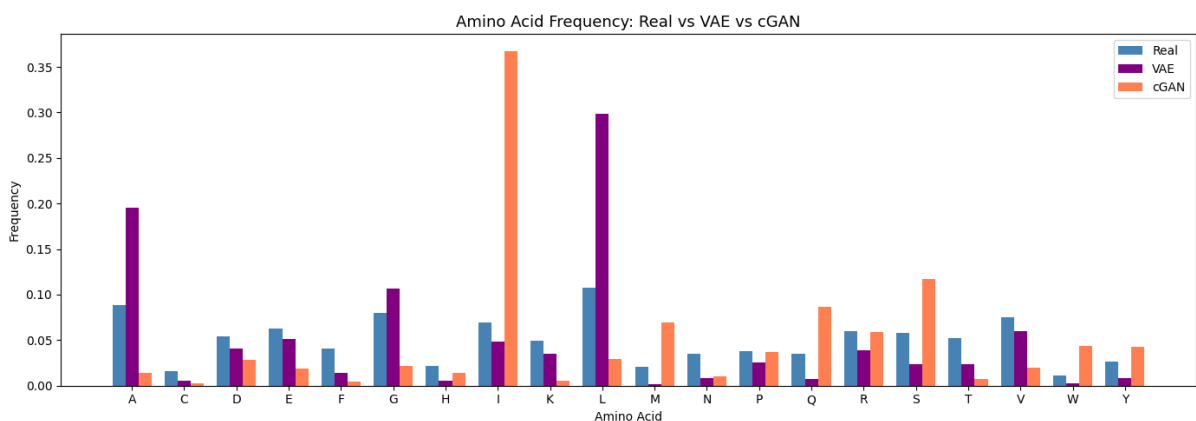


Figure 11: Comparison of amino acid frequency distributions between real sequences and those generated by VAE and cGAN models. The generative models approximate the real distribution with some deviations in specific residues.

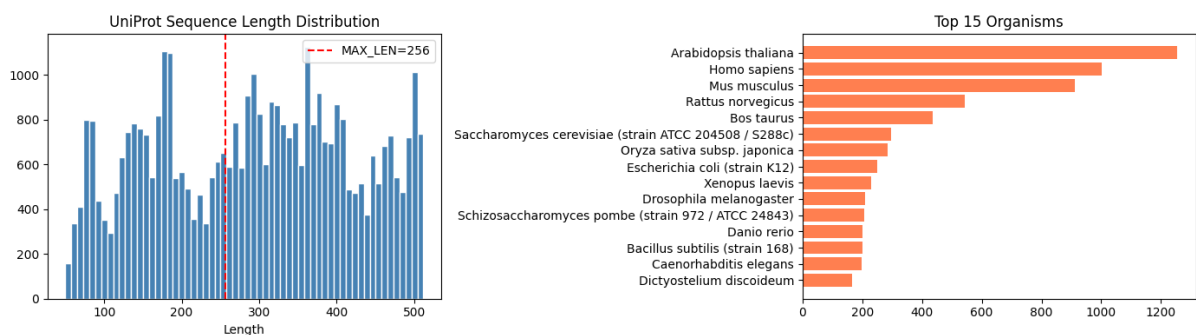


Figure 12: (Left) Distribution of UniProt sequence lengths with a truncation threshold at 256 residues. (Right) Top 15 organisms represented in the dataset, highlighting species diversity.

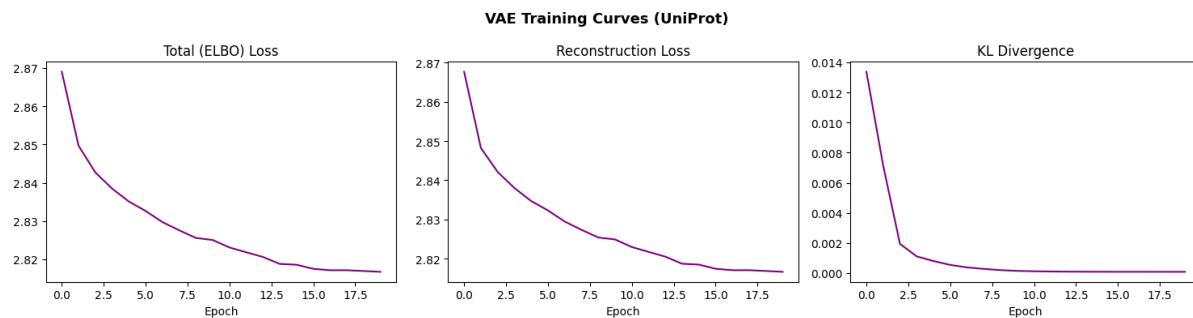


Figure 13: Training curves of the VAE model on the UniProt dataset, showing ELBO loss (left), reconstruction loss (middle), and KL divergence (right). The smooth convergence indicates stable optimization.

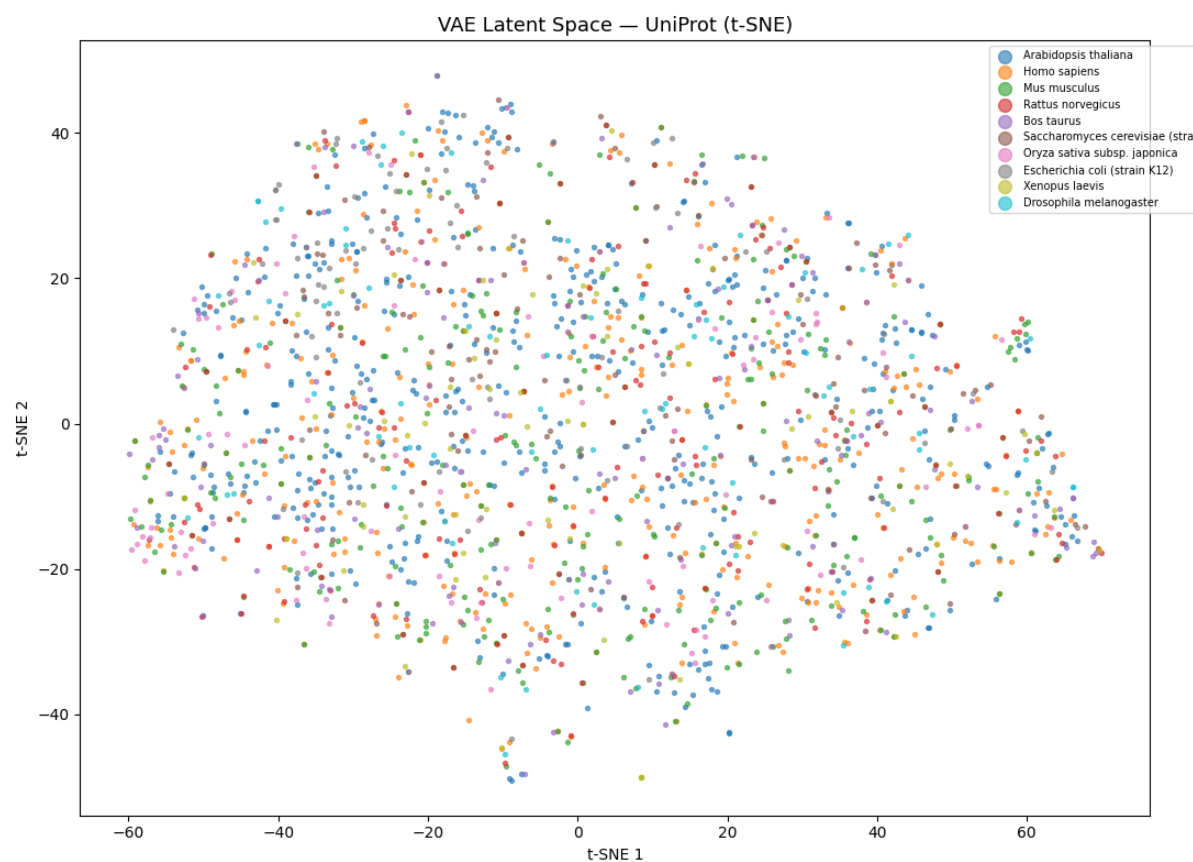


Figure 14: t-SNE projection of the VAE latent space for UniProt sequences. The dispersed structure suggests a more complex and diverse feature distribution compared to dataset.

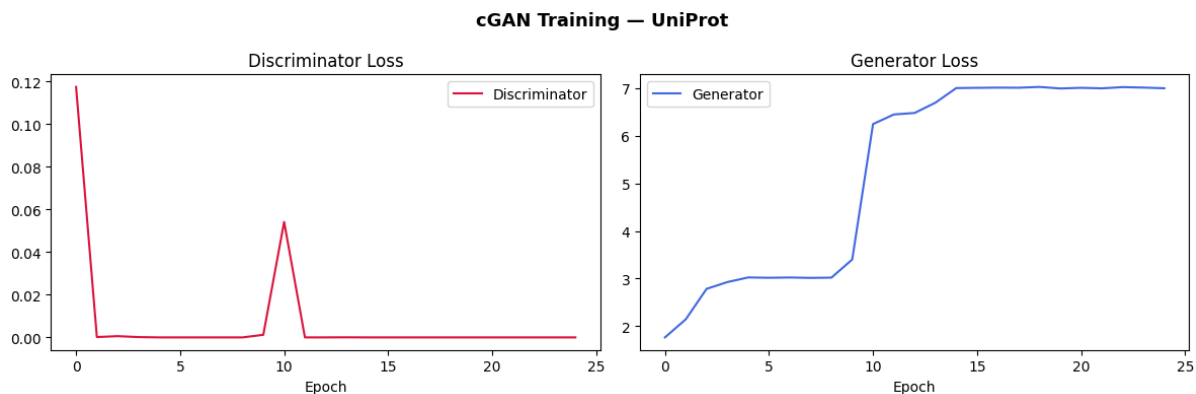


Figure 15: Training dynamics of the cGAN on the UniProt dataset, showing discriminator loss (left) and generator loss (right). The losses indicate stable adversarial learning with gradual generator improvement.

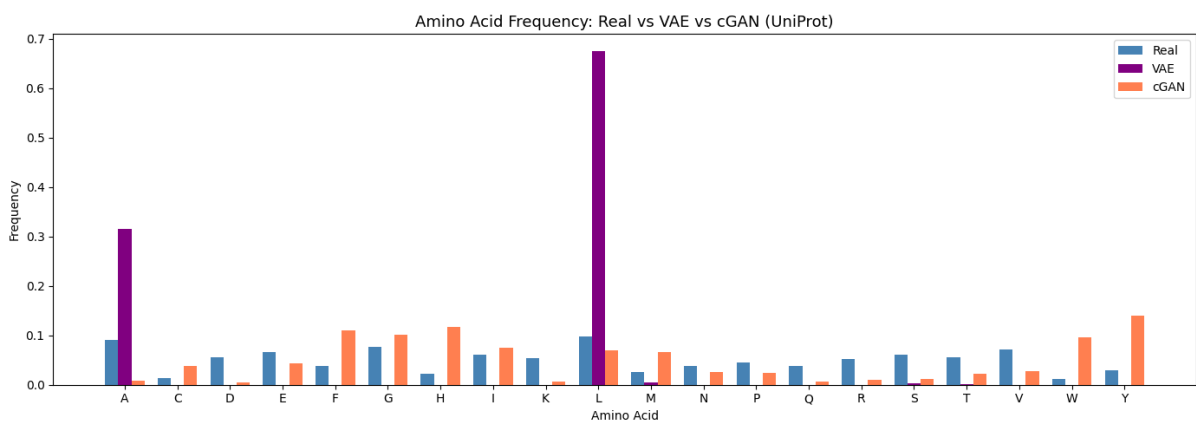


Figure 16: Comparison of amino acid frequency distributions for real UniProt sequences and those generated by VAE and cGAN models. While overall trends are captured, noticeable differences remain in specific amino acids.

8 Comparative Summary

Table 23: ISA-1 — Model Comparison (Test Set)

Model	Params	Accuracy	Macro-F1	Epochs	Converges
1D-CNN	656K	99.94%	0.9994	5	Fast (Ep.2)
LSTM	2,375K	99.74%	0.9972	7	Slow (Ep.4)
GRU	1,816K	99.94%	0.9993	7	Medium (Ep.4)

Table 24: ISA-2 — Generative Model Comparison (Pfam Dataset)

Model	Params	Unique	AA Entropy	Latent Structure
VAE	2,492K	200/200 (100%)	2.287	Excellent (family clusters)
cGAN	13,033K	200/200 (100%)	2.289	N/A (no encoder)

9 Discussion

9.1 Classification Results

The near-perfect accuracy ($> 99.7\%$) across all three architectures demonstrates that the amino acid sequence alone carries sufficient information to identify protein family membership. The **1D-CNN's dominance** with fewer parameters is likely due to its ability to capture short conserved sequence motifs (functional domains, active site signatures), which are the primary discriminating features among Pfam families. The recurrent models (LSTM/GRU), while capable of capturing long-range dependencies, may be over-parameterised for this task.

9.2 Generative Model Limitations

VAE: The Pfam VAE is successful and yields meaningful latent geometry. The UniProt VAE suffers from posterior collapse, a known challenge for sequence VAEs. Solutions include: (1) δ -VAE or free-bits regularisation, (2) replacing the GRU decoder with a Transformer, (3) using VQ-VAE, or (4) training with a much stronger KL annealing schedule.

cGAN: The discriminator-wins failure mode is a fundamental challenge when the discriminator operates on discrete sequences. The argmax operation is non-differentiable, breaking the gradient flow from discriminator to generator. Standard solutions include using Gumbel-Softmax relaxations, WGAN-GP, or training on soft (probability) representations rather than discrete tokens.

9.3 UniProt cGAN Generated Sequences

Despite training instability, the UniProt cGAN generates amino-acid-diverse sequences rich in hydrophobic and aromatic residues (Y, H, W, F, G, I) — patterns consistent with transmembrane proteins and hydrophobic cores, which are prevalent in the Swiss-Prot database. The organism conditioning appears to have limited biological fidelity, as expected given the discriminator collapse.

10 Limitations and Future Work

1. **Dataset scale:** Only 20 families (28K sequences) were used for classification. Scaling to all Pfam families ($>16,000$) would better test model robustness.
2. **VAE posterior collapse:** Replace the GRU decoder with a Transformer decoder and use free-bits regularisation to prevent KL collapse on larger datasets.
3. **GAN training stability:** Implement WGAN-GP with gradient penalty, or switch to a Gumbel-Softmax GAN to enable backpropagation through discrete tokens.
4. **Biological validity:** Evaluate generated sequences using bioinformatics tools such as BLAST (sequence similarity), ESMFold (predicted structure plausibility), or Rosetta (energy scoring).
5. **Pre-trained models:** Leverage ESM-2 or ProtTrans embeddings as input features instead of one-hot/embedding-from-scratch, which would likely yield stronger

representations.

6. **Sequence diversity metrics:** Add Levenshtein distance-based diversity, novelty against training set, and biochemical property distributions as additional evaluation metrics.

11 Conclusion

This project demonstrates the power of deep learning for protein sequence analysis. For **ISA-1 (classification)**, all three architectures (1D-CNN, LSTM, GRU) achieved exceptional accuracy exceeding 99.7% on the 20-family Pfam test set, with the 1D-CNN emerging as the most efficient model (99.94% accuracy, 656K parameters, convergence in 2 epochs). For **ISA-2 (generation)**, the VAE successfully learned a structured, biologically meaningful latent space on the Pfam dataset, enabling smooth interpolation between protein families. The cGAN generated fully unique, amino-acid-diverse sequences conditioned on protein family or organism identity. Both generative models face challenges on larger, less curated datasets (UniProt), with the VAE suffering posterior collapse and the cGAN facing discriminator over-fitting. These represent clear directions for future architectural improvements.

Code availability: All experiments were conducted in Google Colab with PyTorch. Saved models: `CNN1D.best.pt`, `LSTM.best.pt`, `GRU.best.pt`, `vae.best.pt`, `cgan.G.pt`, `cgan.D.pt`, `vae.uniprot.pt`, `cgan.G.uniprot.pt`.

References

- [1] El-Gebali, S. et al. (2019). The Pfam protein families database in 2019. *Nucleic Acids Research*, 47(D1), D427–D432.
- [2] UniProt Consortium (2023). UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*, 51(D1), D523–D531.
- [3] Kingma, D.P. & Welling, M. (2014). Auto-Encoding Variational Bayes. *ICLR 2014*.
- [4] Mirza, M. & Osindero, S. (2014). Conditional Generative Adversarial Nets. *arXiv:1411.1784*.
- [5] Hochreiter, S. & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), 1735–1780.
- [6] Cho, K. et al. (2014). Learning Phrase Representations using RNN Encoder–Decoder. *EMNLP 2014*.
- [7] Romero, P.A. & Arnold, F.H. (2009). Exploring protein fitness landscapes by directed evolution. *Nature Reviews Molecular Cell Biology*, 10(12), 866–876.
- [8] Lin, Z. et al. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637), 1123–1130.
- [9] Higgins, I. et al. (2017). beta-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework. *ICLR 2017*.

- [10] Lim, J. et al. (2018). Molecular generative model based on conditional variational autoencoder for de novo molecular design. *Journal of Cheminformatics*, 10(1), 1–9.